STRAND 1: INTRODUCTION TO COMPUTING

Lesson 01: Next-Generation Computers and Supercomputing

Sub-strand 1: Components of Computers and Computer Systems | Indicator: B9.1.1.1.1

Standard: International British English	**Target Level:** Basic 9 (Grade 9)	**Pacing:** Day 1 of 40 (Part 1)
Curriculum Ref: NaCCA B9.1.1.1	**Learning Mode:** Theory Q&A; + Visuals	**Assessment:** Checkpoint Drill

Part 1: Foundational Preamble and Basic Concepts

Before examining advanced next-generation computers, it is essential to understand foundational computing concepts and how electronic computers evolved across generations.

Q1. What is a computer?

Ans. A computer is an electronic device that accepts raw data as input, processes it according to stored instructions, and produces meaningful information as output.

Q2. What are the five historical generations of electronic computers?

Ans. The five historical computer generations are:

- **First Generation (1940–1956):** Used vacuum tubes for circuitry and magnetic drums for memory.
- **Second Generation (1956–1963):** Replaced vacuum tubes with smaller and more reliable transistors.
- **Third Generation (1964–1971):** Utilised integrated circuits (ICs) combining many transistors on silicon chips.
- **Fourth Generation (1971–Present):** Built using microprocessors that contain thousands or millions of integrated circuits on a single chip.
- **Fifth Generation (Present & Beyond):** Focused on artificial intelligence, parallel processing, and voice recognition.

Q3. What is a classical computer bit?

Ans. A bit, short for binary digit, is the most basic unit of data in classical computing and can only hold a value of either 0 or 1 at any single moment.

Q4. How does a classical microprocessor process information?

Ans. A classical microprocessor processes data sequentially by switching billions of tiny electronic transistors on (representing 1) or off (representing 0).

Q5. What is the main physical limitation facing classical microprocessors today?

Ans. Classical microprocessors are reaching physical limits because transistors cannot be made much smaller without generating excessive heat and experiencing quantum leakage.

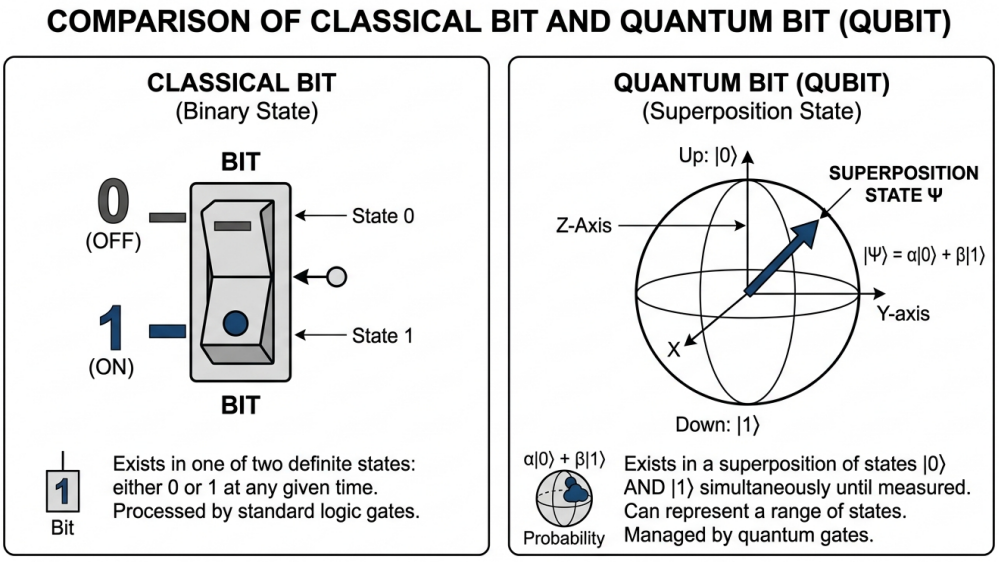


Figure 1.1: Structural Comparison between a Classical Binary Bit (0 or 1) and a Quantum Qubit in Superposition.

Part 2: Trends in Next-Generation Computers

Q6. What is a next-generation computer?

Ans. A next-generation computer is an advanced computing system designed using emerging technologies such as quantum physics, optical light circuits, and artificial intelligence to surpass the processing boundaries of traditional silicon computers.

Q7. State five major features expected to be seen in next-generation computers.

Ans. Five key features expected in next-generation computers are:

- **Unprecedented Processing Speed:** Ability to perform complex mathematical calculations millions of times faster than modern desktop computers.
- **Nanoscale and Molecular Architecture:** Use of atomic-scale components, carbon nanotubes, or optical photons instead of standard silicon transistors.
- **Embedded Artificial Intelligence:** Hardware built specifically to execute neural network routines and machine learning models autonomously.
- **Minimal Energy Consumption:** Highly efficient power consumption with innovative low-resistance or superconducting materials.
- **Quantum State Manipulation:** Processing data using subatomic particles that can exist in multiple states at the same time.

Q8. What is an optical computer?

Ans. An optical computer is a computing machine that uses beams of light (photons) instead of electrical currents (electrons) to process and transmit data at higher speeds with lower heat.

Part 3: Quantum Computing Foundations

Q9. What is quantum computing?

Ans. Quantum computing is a rapidly emerging branch of computer science that harnesses the unique laws of quantum mechanics to solve problems that are too complex for classical computers.

Q10. What is a qubit?

Ans. A qubit, or quantum bit, is the basic unit of information in a quantum computer that can represent 0, 1, or any proportion of both simultaneously.

Q11. What is superposition in quantum computing?

Ans. Superposition is a quantum property that allows a qubit to exist in multiple states at once rather than being confined to a single definite state of 0 or 1.

Q12. What is quantum entanglement?

Ans. Quantum entanglement is a phenomenon where two or more qubits become inextricably linked so that the physical state of one instantly dictates the state of another, regardless of distance.

Q13. Why can a quantum computer process exponentially more data than a classical computer?

Ans. A quantum computer processes data exponentially because each additional qubit doubles the total number of computational states that can be analysed simultaneously.

Part 4: Google Sycamore and Classical Supercomputers

Q14. What is a classical supercomputer?

Ans. A classical supercomputer is an exceptionally powerful machine composed of thousands of linked traditional processors working together to solve massive data-intensive problems.

Q15. Give one prominent real-world example of a classical supercomputer.

Ans. The IBM Summit supercomputer at the Oak Ridge National Laboratory is a leading example of a classical supercomputer.

Q16. What is Google Sycamore?

Ans. Google Sycamore is an advanced quantum processor developed by Google that operates using superconducting circuits kept at cryogenic temperatures near absolute zero.

Q17. How many physical qubits were built into the Google Sycamore processor?

Ans. The Google Sycamore processor was fabricated with fifty-four (54) physical qubits arranged in a rectangular grid.

Q18. How many qubits were operational during Google Sycamore's milestone demonstration?

Ans. During its milestone experiment, fifty-three (53) qubits were fully operational because one qubit did not function properly.

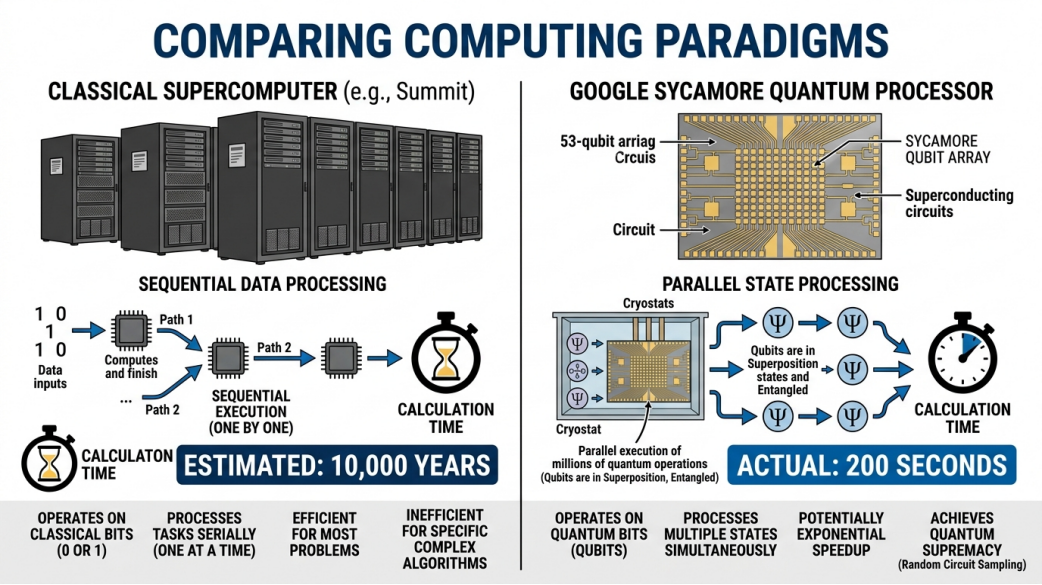


Figure 1.2: Architectural Paradigm Comparison: Classical Supercomputer vs Google Sycamore Quantum Processor.

Part 5: Processing Power Comparison and Quantum Supremacy

Q19. What specific benchmark task did Google use to test the Sycamore processor?

Ans. Google tasked Sycamore with sampling the output distribution of a random quantum circuit, which tests how quickly a machine can verify complex random patterns.

Q20. How long did it take Google Sycamore to complete this benchmark task?

Ans. Google Sycamore completed the computational task in approximately two hundred (200) seconds.

Q21. How long would the world's fastest classical supercomputer take to complete the exact same task?

Ans. Google estimated that the fastest classical supercomputer at the time would take approximately ten thousand (10,000) years to calculate the same result.

Q22. What is quantum supremacy?

Ans. Quantum supremacy, also known as quantum advantage, is the technical milestone where a programmable quantum device successfully solves a calculation that no classical computer can complete in any feasible amount of time.

Q23. Mention four real-world fields that will benefit from next-generation supercomputing.

Ans. Four fields that will benefit from next-generation supercomputing are:

- **Pharmaceutical Medicine Design:** Simulating molecular structures to discover lifesaving drugs rapidly.
- **Climate and Weather Modelling:** Forecasting extreme atmospheric conditions with exceptional precision.
- **Cryptography and Data Security:** Developing uncrackable encryption protocols to protect global communication networks.
- **Materials Science:** Designing lighter, stronger, and more energy-efficient materials for engineering and aerospace.

Key Terminology and Definitions

Term	Meaning in Plain English	Context / British Note
Bit	A single classical digit holding value 0 or 1.	Binary digit in classical computing.
Qubit	A quantum bit capable of existing in superposition.	Basic unit of quantum information.
Superposition	Ability to represent multiple states simultaneously.	Governed by quantum probability.
Entanglement	Physical linking of particles across distances.	Enables instant coordination.
Cryogenic	Operating at temperatures close to absolute zero.	Required for Sycamore stability.
Quantum Supremacy	Quantum computer solving problems classical ones cannot.	Also called quantum advantage.

Day 1 Assessment: Checkpoint Mastery Drill

Complete these questions to confirm your understanding of Lesson 01. Review the marking criteria below each question.

Question 1: State the main difference between a classical bit and a quantum qubit.

Marking Guide: A classical bit can only be in one state (either 0 or 1) at a time, whereas a quantum qubit can represent both 0 and 1 simultaneously through superposition.

Question 2: How many physical qubits are contained on the Google Sycamore chip, and how many were operational during its landmark test?

Marking Guide: Google Sycamore contains 54 physical qubits, of which 53 were operational during the landmark experiment.

Question 3: State the time difference between Google Sycamore and the classical supercomputer IBM Summit when performing the random circuit sampling task.

Marking Guide: Google Sycamore completed the task in 200 seconds, whereas IBM Summit was estimated to require approximately 10,000 years.

Question 4: Mention three characteristics expected in next-generation computer systems.

Marking Guide: Any three of: Extreme processing speed, nanoscale/molecular components, embedded artificial intelligence, ultra-low energy consumption, quantum state processing.

Question 5: Define the term *quantum supremacy*.

Marking Guide: Quantum supremacy is the milestone where a quantum computer solves a problem that no classical supercomputer can complete within a feasible timeframe.